INCH-POUND

MIL-R-50664B (AR)
26 December 1990
SUPERSEDING
MIL-R-50664A (AR)
15 April 1971

MILITARY SPECIFICATION

REGULATOR, 29 VOLT, BOARD ASSEMBLY: 10548961

This specification is approved for use within the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 29-volt regulator board assembly, hereinafter referred to as a 29-volt regulator, used in the regulator assemblies of Power Supply PP-4812/VPS-2, part of Radar Set AN/VPS-2 and in Subassembly PP-6324/TPM-22, part of Radar Test Set AN/TPM-22 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection of, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548822 - Inductor, power
10548932 - Transformer, Saturable Core
10548961 - 29-Volt Regulator Board Assembly

PACKAGING DATA SHEET

PS1054896I - Packaging **of 29-Volt Regulator**

(Copies of specifications, standards, handbooks, drawings publications and other Government documents required by contractors in **connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)**

2.2 Order of Precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. **REQUIREMENTS**

3.1 Materials. Materials shall be in accordance with the **applicable drawings and specifications.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548961, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the 29-volt regulator shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3*1 Vibration. The 29-volt regulator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as *follows:*

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Shock. The 29-volt regulator shall not suffer damage and shall meet requirements of **3.4 and 3.5 after exposure to** a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 OPerating. The 29-volt regulator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-oPerating. The 29-volt regulator shall meet the requirements of **3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).**

3.3.4 **Temperature.**

3.3.4.1 Operating. The 29-volt regulator shall meet the requirements o f 3.4 and 3.5 when stabilized thermally at any temperature between -4o deg.F and +155 deg.F.

Source: https://assist.dla.mil -- Downloaded: 2026t0-22T20:33Z
Check the source to verify that this is the currentversion before use.

3.3.4.2 Non-operating (storage) . The 29-volt regulator shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg.F for a period of 24 hours and +160 deg.F for a period of 4 hours (after temperature stabilization) , and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The 29-volt regulator shall meet the requirements of 3.4 and 3.5; after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the 29-volt regulator board assembly.

3.5 Performance. The 29-volt regulator shall produce results as given in 3.5.1 when supplied with the input voltages specified in Table I (Input characteristics).

TABLE I. Input characteristics.

REQUIREMENT	CHARACTERISTIC
Input voltage 1/	36 to 50 Vdc; regulated to ±0.5%
Input signal 1/	22 ± 1 **KHz, 71.5 ± 12.5V peak-to-peak** square wave

1/ Input voltages and signals are connected into the circuit using test equipment as specified in 4.6.2.3 and test fixture setup as specified in 4.7.3.

3.5.1 Output terminals. The output terminals are E5 (-) and E6 (+).

3.5.1.1 Output voltage level potentiometer R2. R2 shall control the output voltage between circuit board **terminals E5 (-)** and E6 (+). The nominal-output voltage shall be 29.30 ± 0.30 volts (V) with a load current of 10.0 + 0.2 amperes direct current (Adc).

3.5.1.2 **Current limit potentiometer R17. R17 shall control the point at which the current limiting circuitry becomes** effective. With a short applied between E5 and E6, maximum current drain shall be controlled by R17 and shall not exceed 16.0 Adc at **room ambient and 17.5 Adc at specified temperature extremes. The output shall automatically recover after removal of the short circuit.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

3.5.1.3 Voltage regulation, line. The output voltage shall not vary **more than** ±0.60 volt direct current (Vdc) with an input voltage variation from 36 to 50 Vdc and a constant load current of **10.0** A.

3.5.1.4 Voltage regulation, load. The output voltage shall not vary more than ±0.60 Vdc with the input voltage fixed at 43 ± 7 Vdc when the output current varies from 3.0 to 10.0 A.

3.5.1.5 Output ripple. The ripple voltage shall not exceed 0.5 V peak-to-peak when the input voltage is 43 ± 0.43 Vdc and the load current is 10.0 A. Exclude high-frequency spikes which occur at approximately 25 usec intervals.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task)F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor **may use his own or** any **other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed** necessary to assure supplies and services-conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5 and all applicable drawings. The **inspection set forth in this specification shall become a part of** the **contractor's overall inspection system** or **quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance** does **not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.**

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of **the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

	Temperature	+50°F to +90°F
b	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) 29-volt regulators for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table II (Functional tests) and Table III (Special sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 **Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any assembly, component or test specimen to comply with any of the stated requirements.**

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which **a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

4.4.2 Examination and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies, and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	**None**	**N/A**
Major:		
101. **Electrical**	**3*4**	4.7.2
102. Performance	**3*5**	4.7.3
Minor:	**None**	N/A

4.5 Special sampling. One 29-volt regulator out of the first **50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and** tests **of** Table II and shall then meet the requirements and tests in Table 111. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

TABLE III. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply **and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.**

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the **'Methods** of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and 'Test **Facilities" requirements** of **MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance** of **test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table IV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IV. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Signal generator	Capable of producing a 22 ± 1 KHz, 71.5 ± 12.5 V peak-to-peak square wave. Output impedance less than 20 ohms.
2.	Power supply	DC, 36 to 50V, 20 A, regulated to ±0.5%.
3.	Voltmeter	DC, 0 to 50 V, 0.01% accuracy, 20,000 ohms/volt or greater.
4.	Oscilloscope	Input sensitivity: 20 mV/cm. Input impedance: 1 megohm. Vertical passband: DC to 1 MHz. Horizontal deflection: 5 us/cm.
5.	**Ammeter**	**DC, 0 to 20 A, 1% accuracy.**
6.	**Test fixture**	As specified in Figure 1. See Table V for fixture parts list.

TABLE V. Test fixture parts list.

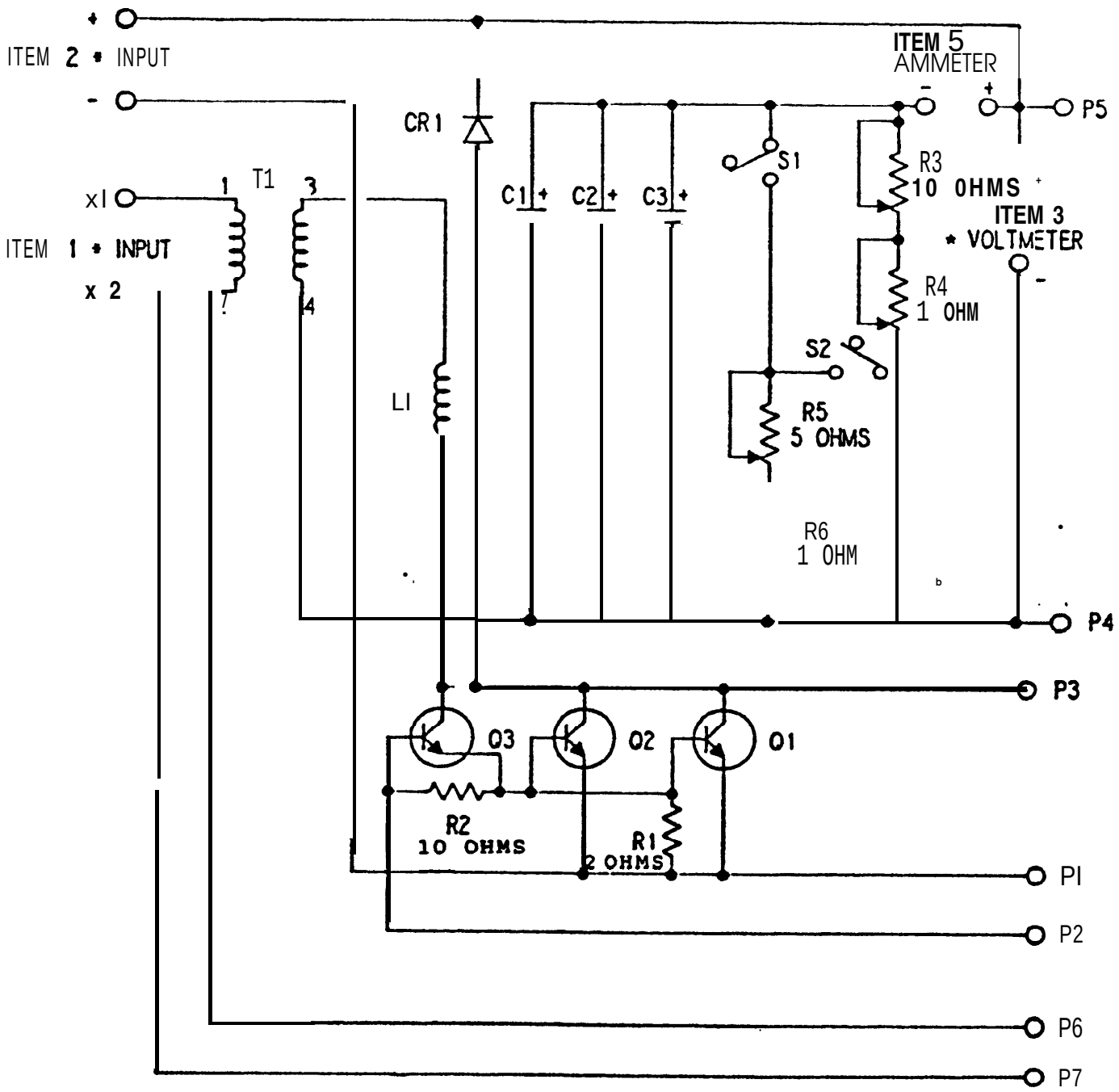
ITEM	NOMENCLATURE	DESCRIPTION
C1,C2,C3	Capacitor	140 uF, 40V
CR1	Diode	JAN 1N3891
L1	Inductor	Power, 1.0 mH, 12A, 10548822
Q1,Q2,Q3	Transistor	Power, NPN silicon, 2N5038
R1	Resistor	Fixed 2.0 ohms, WW, power, RWR89S2R00FM
R2	Resistor	Fixed 10 ohms, film, RLR20C100JM
R3	Resistor	Variable 0 to 10 ohms, 100W
R4	Resistor	Variable 0 to 1 ohm, 10 W
R5	Resistor	Variable 0 to 5 ohms, 500W

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

TABLE V. Test fixture parts list. (continued)

ITEM	NOMENCLATURE	DESCRIPTION
R6	Resistor	Variable 0 to 1 ohm, 250 W
S1	Switch	SPST, general purpose, 20 A rated
S2	Switch	SPST, general purpose, 20 A rated
T1	Transformer	Saturable core, 10548932

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.



* Refer to 4.6.2.3

FIGURE 1. Test fixture, regulator board.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category **8,** Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I, only steps 4 through 8 of paragraph II-2.1 apply. The 29-volt regulator shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the 29-volt regulator shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section IX of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 Hertz.

After the application of all the shock impulses, the 29-volt regulator shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810D The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 **with constant temperature extremes as specified in 3*3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.**

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:33Z
Check the source to verify that this is the current version before use.

4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The selected 29-volt regulators shall be tested as follows: All following tests shall be performed with the test equipment connected to the 29-volt regulator as shown in Figure 2.

4.7.3.1 Output voltage level potentiometer R2. Using signal generator (item 1, Table IV), apply a 22 ± 1 kilohertz square wave to test fixture terminals Xl and X2 and adjust item 1 so that the amplitude of the input signal is 71.5 ± 12.5 V peak-to-peak. Using power supply (item **2,** Table IV), apply 43 ± 7 Vdc between the (+) and (-) terminals on the test fixture. With switch S1 closed, S2 open and using voltmeter (item 3, Table IV) and ammeter (item 5, Table IV), verify the voltage between P4 (-) and P5 (+) on the test fixture can be adjusted (using R2) to 29.30 ± 0.30 Vdc with a load current of 10.0 ± 0.2 Adc. Vary resistive load (R3-R6) to obtain current indication of 10.0 ± 0.2 Adc.

4.7.3.2 Current limit Potentiometer R17. Perform the following steps in sequence:

a. **Set up conditions specified in 4.7.3.1.**

b. Adjust R2 for voltage indication of 29.30 ± 0.30 Vdc.

c. Adjust R17 fully CCW.

d. Close load switch S1 (see Figure 1).

e. Close short switch S2 (see Figure 1).

f. Adjust R17 CW until output current equals 16.0 Adc.

g. Open S2.

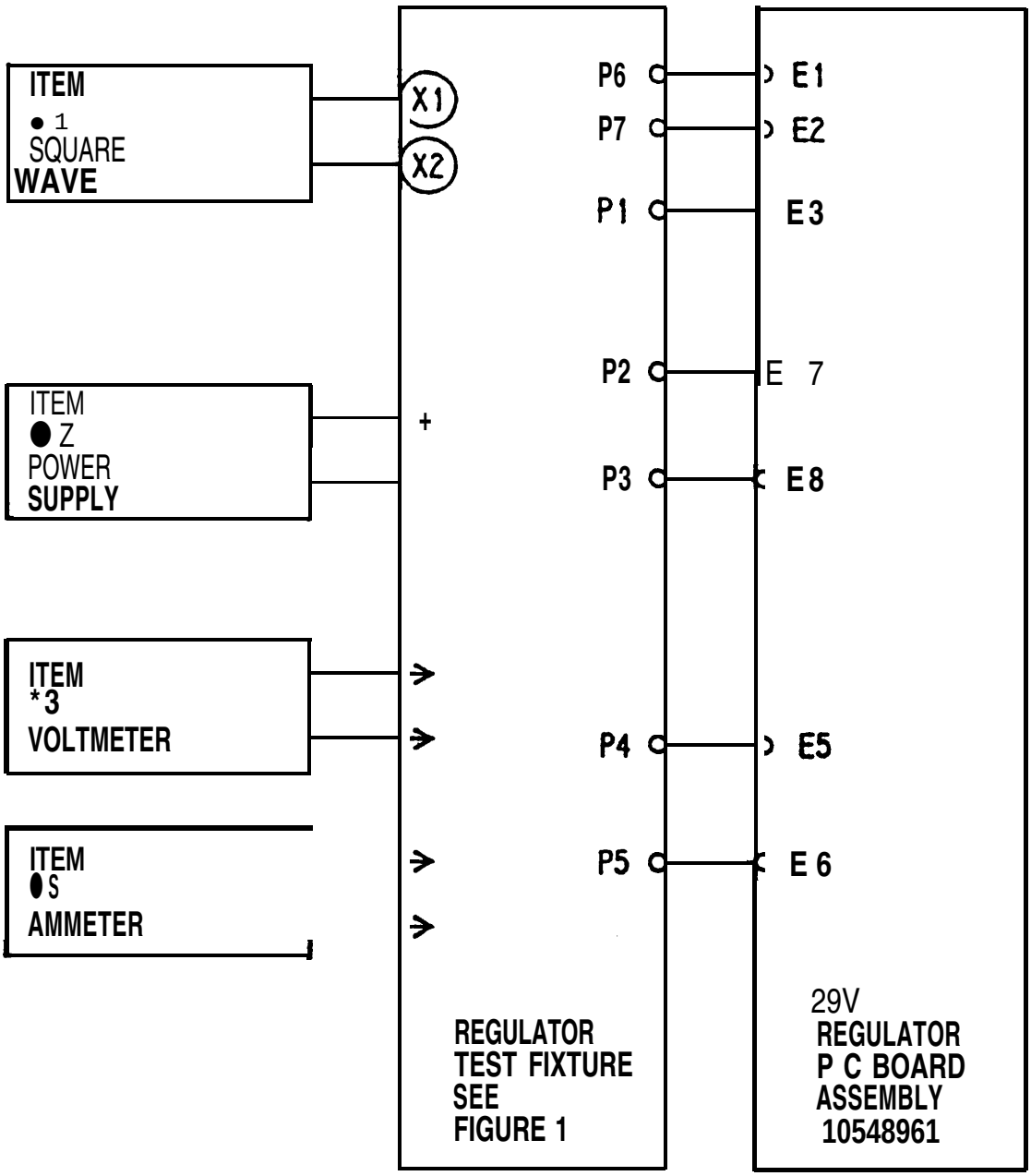
he Vary resistive load (R3-R6) to obtain current indication of 10.0 Adc.

i. Record voltmeter indication using item 3.

j. Close S2 momentarily, then open.

k. Using voltmeter (item 3, Table IV), verify that output **automatically returns to within ±0.30 volts of level in "i" above.**

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:33Z
Check the source to verify that this is the current version before use.



● Refer to 4.6.2.3

FIGURE 2. Test harness setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

4.7.3.3 Voltage regulation, line. With R2 and R17 adjusted as in 4.7.3.2, adjust the resistive load (R3-R6) to obtain 10.0 A. Vary the input voltage (item 2, Table IV) from 36 to 50 Vdc. Verify that the output voltage does not vary more than **±0.60** Vdc.

4.7.3.4 Voltage regulation, load. Set the input voltage to 43 ± 7 vdc. With R2 and R17 adjusted as specified in 4.7.3.2, adjust resistive loads (R3 and R4) to obtain 3.0 A with switch S1 open. Record the voltage indication on item 3. Close switch S1 and vary resistive loads (R5 and R6) obtain 10.0 A. Using voltmeter (item 3, Table IV), verify the output voltage does not vary more than **±0.60** vdc.

4.7.3.5 Output ripple. With the input voltage set at 43 ± 0.43 Vdc and R2 and R17 adjusted as specified in 4.7.3.2, adjust the resistive loads (R3-R6) to 10.0 A. Using oscilloscope (item 4, Table IV), verify the ripple does not exceed 0.50 V peak-to-peak. Exclude high-frequency spikes which occur at approximately 25 usec intervals.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with packaging Data Sheet PS10548961. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 29-volt regulator is used in the regulator assembly part of the Power Supply PP-4812/VPS-2, part of Radar Set AN/VPS-2 and It is also used in Subassembly PP-6324/TPM-22, part of Radar Test Set AN/TPM-22. The 29-volt regulator is **used** to **regulate the output voltage** within the specified requirements for functional operation of Radar Set AN/VPS-2. In the test set, it is used to check voltage regulator performance of the radar power supply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-1, Picatinny Arsenal, NJ **07806-5000.** This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Fire control system
Power Supply PP-4812/VPS-2
Radar Set AN/VPS-2
Radar Test Set AN/TPM-22
Subassembly PP-6324/TPM-22
VADS

6.6 Changes from previous issue. Marginal notations are not **used in** this revision to identify changes with respect to the previous issue **due to the** extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A691)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-50664B (AR)	26 December 1990

3. DOCUMENT TITLE

REGULATOR, 29 VOLT, BOARD ASSEMBLY: 10548961

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

		DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
US Army ARDEC Standardization & Specification Office	(201) 724-6674	880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.